

LAB EXPERIMENT RECORD

Set 1: Monthly Sales Data

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To visualize monthly sales data using a line chart, bar chart, and scatter plot in R, and to build an interactive dashboard combining the line chart and bar chart using R Shiny.

Dataset: Monthly Sales

Month	Sales (in \$)
January	15000
February	18000
March	22000
April	20000
May	23000

Note: The experiment table gives only Month and Sales. Question 3 needs an advertising budget and Question 2 needs yearly product sales, so the R programs use the following values: Advertising Budget (\$) – 2000, 2500, 3200, 2800, 3500 (January to May); Yearly Product Sales (\$) – Product A 120000, Product B 95000, Product C 80000, Product D 65000, Product E 50000.

PROGRAM 1: Line Chart of Monthly Sales

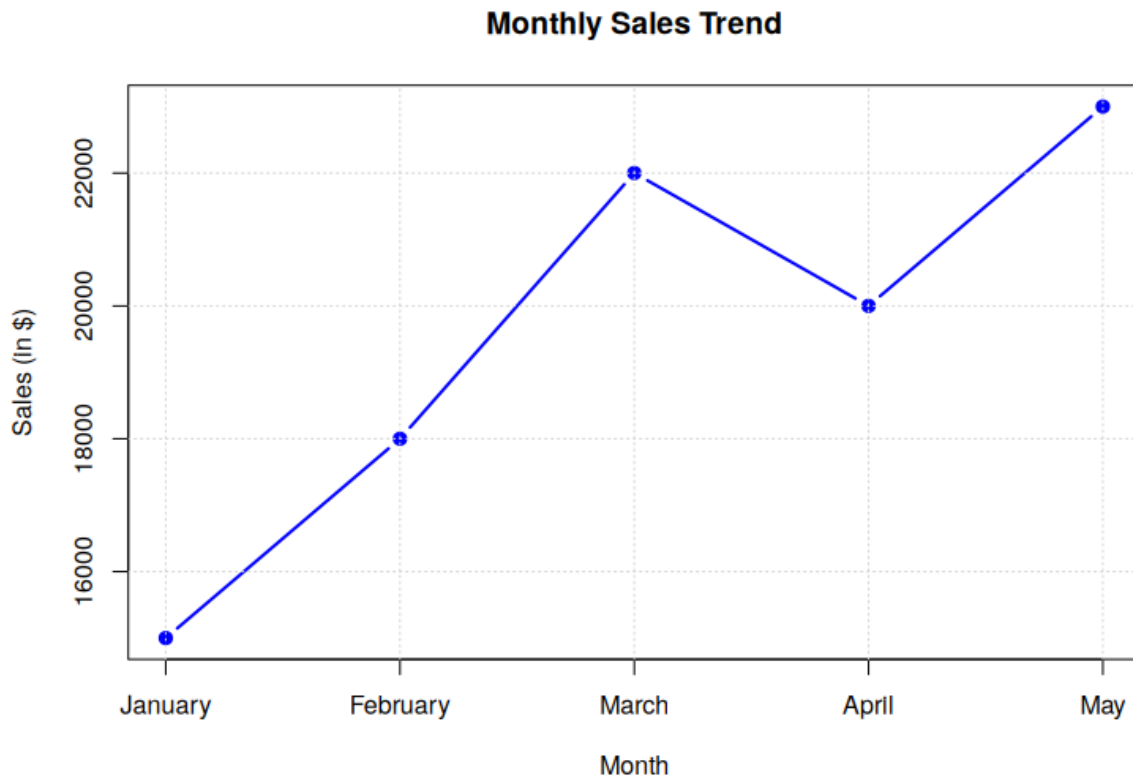
Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

```
# Create the dataset for Set 1
monthly_sales <- data.frame(
  Month = factor(c("January", "February", "March", "April", "May"),
    levels = c("January", "February", "March", "April", "May")),
  Sales = c(15000, 18000, 22000, 20000, 23000),
  Ad_Budget = c(2000, 2500, 3200, 2800, 3500) # Added advertising budget for Q3
)

# View the created dataset
print(monthly_sales)

# 1. Line Chart for Monthly Sales
```

```
plot(as.numeric(monthly_sales$Month), monthly_sales$Sales,  
     type = "b",  
     col = "blue",  
     lwd = 2,  
     pch = 19,  
     xaxt = "n",  
     xlab = "Month",  
     ylab = "Sales (in $)",  
     main = "Monthly Sales Trend")  
axis(1, at = 1:5, labels = levels(monthly_sales$Month))  
grid()
```



Output 1: Monthly Sales Trend

Observation: Sales increase from \$15,000 in January to \$22,000 in March, dip slightly to \$20,000 in April, and reach the highest value of \$23,000 in May. Overall, the sales show an upward trend.

PROGRAM 2: Bar Chart of Top-Selling Products

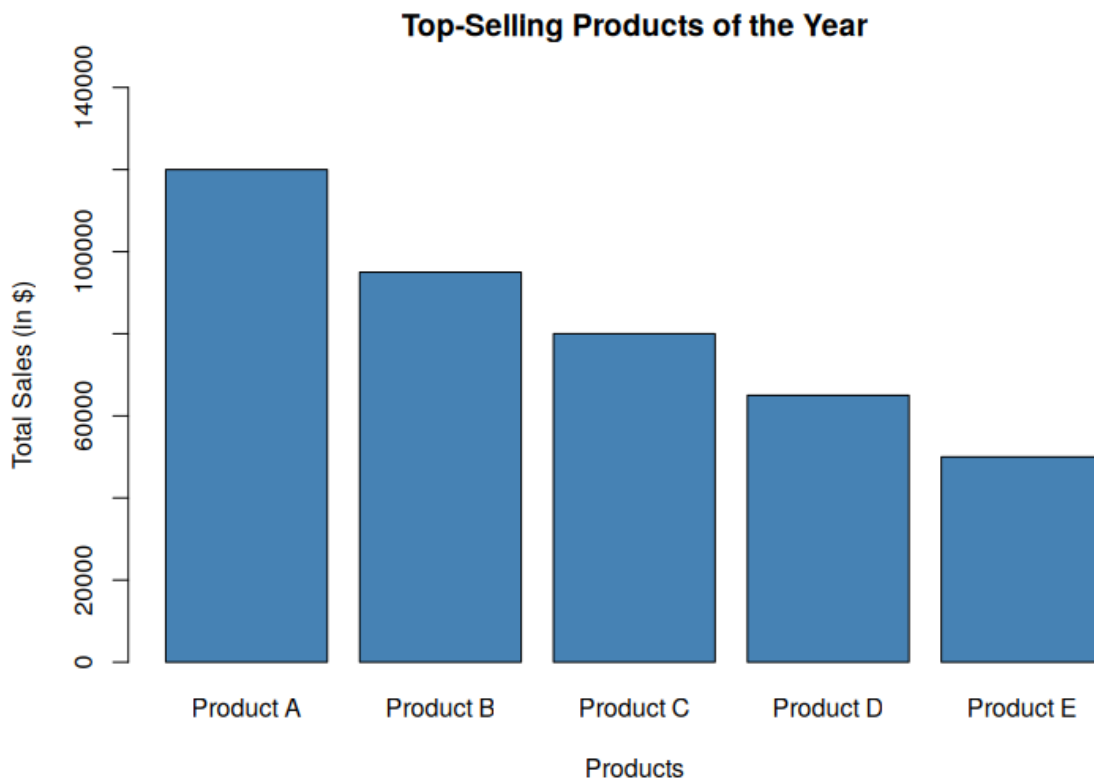
Generate a bar chart to display the top-selling products for the year. Label the chart elements.

```
# Create the dataset for Set 1
monthly_sales <- data.frame(
  Month = factor(c("January", "February", "March", "April", "May"),
    levels = c("January", "February", "March", "April", "May")),
  Sales = c(15000, 18000, 22000, 20000, 23000),
  Ad_Budget = c(2000, 2500, 3200, 2800, 3500) # Added advertising budget for Q3
)

# View the created dataset
print(monthly_sales)

# Sample yearly sales data for top products
product_sales <- c(120000, 95000, 80000, 65000, 50000)
names(product_sales) <- c("Product A", "Product B", "Product C", "Product D", "Product E")

# 2. Bar Chart for Top-Selling Products
barplot(product_sales,
  col = "steelblue",
  xlab = "Products",
  ylab = "Total Sales (in $)",
  main = "Top-Selling Products of the Year",
  ylim = c(0, 140000))
```



Output 2: Top-Selling Products of the Year

Observation: Product A is the top-selling product with \$120,000 in yearly sales, followed by Product B (\$95,000) and Product C (\$80,000). Product E has the lowest sales at \$50,000.

PROGRAM 3: Scatter Plot of Advertising Budget vs Monthly Sales

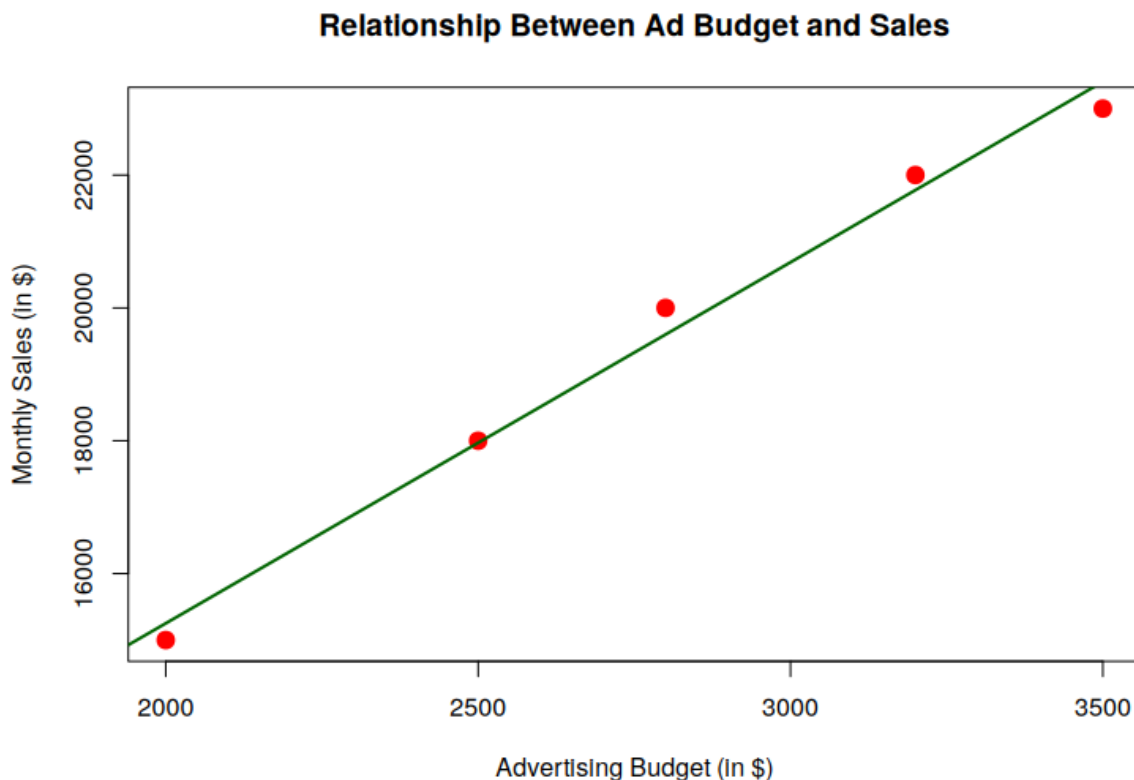
Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

```
# Create the dataset for Set 1
monthly_sales <- data.frame(
  Month = factor(c("January", "February", "March", "April", "May"),
    levels = c("January", "February", "March", "April", "May")),
  Sales = c(15000, 18000, 22000, 20000, 23000),
  Ad_Budget = c(2000, 2500, 3200, 2800, 3500) # Added advertising budget for Q3
)

# View the created dataset
print(monthly_sales)

# 3. Scatter Plot for Ad Budget vs Sales
plot(monthly_sales$Ad_Budget, monthly_sales$Sales,
  col = "red",
  pch = 19,
  cex = 1.5,
  xlab = "Advertising Budget (in $)",
  ylab = "Monthly Sales (in $)",
  main = "Relationship Between Ad Budget and Sales")

# Add a trend line
abline(lm(Sales ~ Ad_Budget, data = monthly_sales), col = "darkgreen", lwd = 2)
```



Output 3: Relationship Between Ad Budget and Sales

Insight: The points lie very close to the upward trend line, showing a strong positive relationship between advertising budget and sales (correlation coefficient about 0.99). Based on the trend line, every additional \$1 spent on advertising is associated with about \$5.43 more in monthly sales. The highest budget (\$3,500 in May) gives the highest sales (\$23,000). With only five months of data, this shows association and not proof of cause.

PROGRAM 4: Interactive Sales Dashboard

Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

```
# install.packages("shiny") # run once if shiny is not installed
library(shiny)

# Create the dataset for Set 1
monthly_sales <- data.frame(
  Month = factor(c("January", "February", "March", "April", "May"),
    levels = c("January", "February", "March", "April", "May")),
  Sales = c(15000, 18000, 22000, 20000, 23000),
  Ad_Budget = c(2000, 2500, 3200, 2800, 3500) # Added advertising budget for Q3
)

# View the created dataset
print(monthly_sales)

# Sample yearly sales data for top products
product_sales <- c(120000, 95000, 80000, 65000, 50000)
names(product_sales) <- c("Product A", "Product B", "Product C", "Product D", "Product E")

ui <- fluidPage(
  titlePanel("Set 1: Interactive Monthly Sales Dashboard"),
  sidebarLayout(
    sidebarPanel(
      sliderInput("sales_filter",
        "Minimum Sales Filter ($):",
        min = 10000,
        max = 25000,
        value = 15000,
        step = 1000)
    ),
    mainPanel(
      plotOutput("linePlot", height = "320px"),
      plotOutput("barPlot", height = "320px")
    )
  )
)

server <- function(input, output) {
  filtered_data <- reactive({
    subset(monthly_sales, Sales >= input$sales_filter)
  })

  output$linePlot <- renderPlot({
    data <- filtered_data()
    plot(seq_len(nrow(data)), data$Sales, type = "b", col = "blue", lwd = 2, pch = 19,
      xaxt = "n", xlab = "Month", ylab = "Sales ($)", main = "Monthly Sales Trend")
    axis(1, at = seq_len(nrow(data)), labels = as.character(data$Month))
    grid()
  }, width = 620, height = 320, res = 96)

  output$barPlot <- renderPlot({
    barplot(product_sales, col = "steelblue",
      xlab = "Products", ylab = "Sales ($)",
```

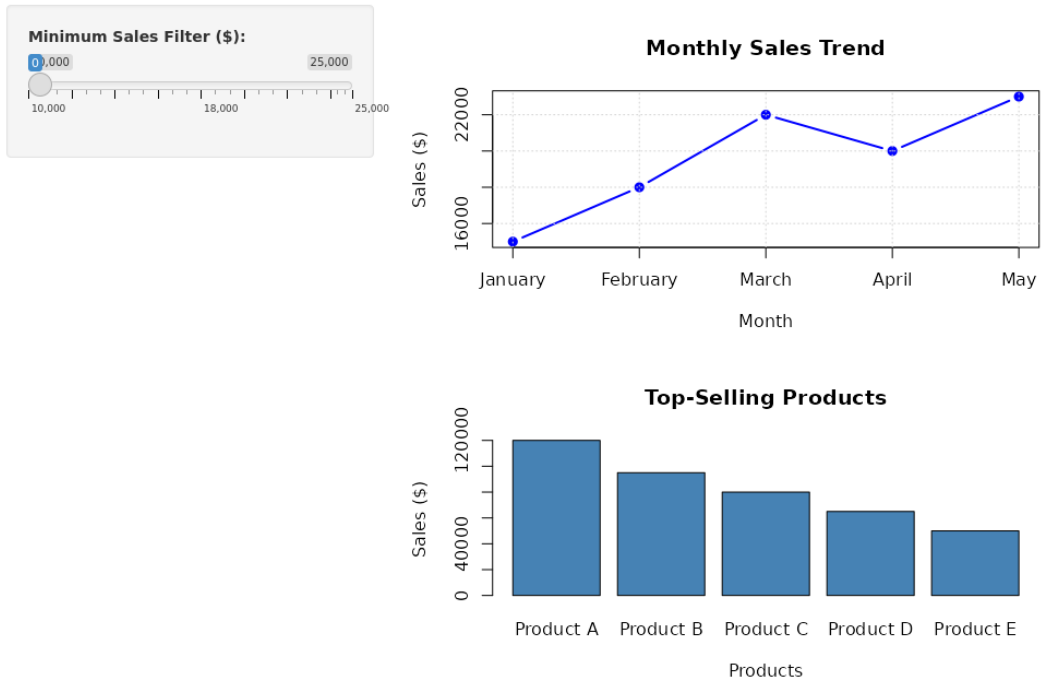
```

    main = "Top-Selling Products")
  }, width = 620, height = 320, res = 96)
}

shinyApp(ui = ui, server = server)

```

Set 1: Interactive Monthly Sales Dashboard



Output 4: Interactive Monthly Sales Dashboard

Observation: The dashboard shows the monthly sales trend and the top-selling products together. Moving the Minimum Sales Filter slider removes the months whose sales are below the selected value from the line chart, so users can focus on the higher-sales months.

Procedure

1. Create the monthly sales data frame with Month, Sales, and Advertising Budget.
2. Use `plot()` with `type = "b"` and `axis()` to draw the line chart with month labels.
3. Create the yearly product sales vector and use `barplot()` to draw the bar chart.
4. Use `plot()` for the scatter plot and `abline()` with `lm()` to add a trend line.
5. Install and load the Shiny package, then create the UI with a slider and two plot outputs.
6. Write the server function with `reactive()` filtering and `renderPlot()`, and run the app using `shinyApp()`.

Result

The monthly sales data was successfully visualized using a line chart, bar chart, and scatter plot in R, and an interactive Shiny dashboard combining the line chart and bar chart was created.

Key R Functions Used

Function	Purpose
<code>data.frame()</code>	Creates the monthly sales dataset.
<code>plot()</code>	Creates the line chart and scatter plot.
<code>axis()</code>	Adds month labels to the x-axis.
<code>barplot()</code>	Creates the bar chart of product sales.
<code>abline()</code> / <code>lm()</code>	Adds a linear trend line to the scatter plot.
<code>fluidPage()</code> / <code>sliderInput()</code>	Creates the Shiny page layout and slider control.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws plots when the slider changes.
<code>shinyApp()</code>	Runs the interactive Shiny application.